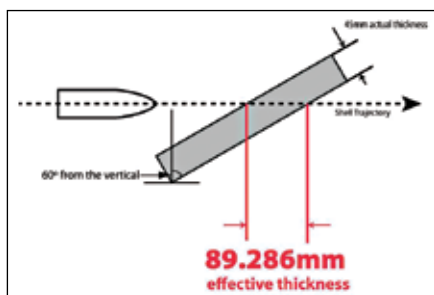


Modern armour is no longer specific to a particular platform because most of it has been designed as modular blocks. It's perfectly feasible to equip even decades old military vehicles with blocks of ERA and make it almost impenetrable to the enemy.

Here's a quick guide to the evolution in arms and armour over the decades.

ARMOUR

- **Plate metal:** Early armoured vehicle were outfitted with plates of metal as a form of armour. It was probably the Germans who discovered (reports are a bit hazy on this) that sloping the armour greatly increased its rating, further increasing the effectiveness of the armour.
- **Composite / Chobham armour:** The increasing effectiveness of armour-piercing shells meant simple plate armour would have to be inordinately



Sloped armour

thick to provide adequate protection. Composite armour is a mix of a very tough ceramic compound sandwiched between layers of metal. The ceramic is hard enough to force most incoming projectiles to shatter on impact.

- **Spaced armour:** This type of armour consists of multiple armour plates that are separated by an air gap, as this Stryker IFV proudly demonstrates. HEAT and HESH rounds explode on contact, doing no damage. It's also effective against RPGs.



That grill you see is spaced armour

- **Reactive Armour:** This is the most advanced type of armour in service today. One form of this armour is Explosive Reactive Armour, where the layers of composite armour are sandwiched together by a layer of explosive. This explosive layer violently displaces the plates when compressed, successfully dissipating the bulk of the energy of the incoming projectile. The British are working on a form of charged explosive reactive armour. Essentially, an electric charge that's carried by the armour module itself triggers the fuse on an incoming projectile, prematurely detonating it.

AMMUNITION

- **Armour Piercing:** Think of this as a rifle bullet, except that it's of a larger calibre and fired from a tank. Plain Jane AP rounds are basically a single block of metal designed with the sole aim of puncturing enemy armour.
- **APCR:** Armour Piercing Composite Rigid ammo was the first step in the evolution of the AP round. This type of ammo is essentially an AP shell with a high-density core, usually made of tungsten. The higher mass gives the shell greater inertia, lending it greater penetrating power.
- **APDS/APFSDS:** This is the most advanced type of AP round used by the military today. The round is called an Armour Piercing Fin Stabilized Discarding Sabot or just APDS if it doesn't

have fins. This round is essentially a slim metallic rod that's encased in a sabot and fired from a gun. This thin metallic rod however, is made from the densest materials known to man and the US uses depleted Uranium. The kinetic energy of this round is immense, punching through most armour with ease.

- **HE:** This shell is primarily used to take down soft targets—troops and trucks. It's basically a high explosive charge that's encased in a relatively thin shell and is designed to explode on contact.
- **HEAT:** High Explosive Anti-Tank shells are a very innovative type of shell used against armour. HEAT rounds don't rely on kinetic energy to penetrate armour. Instead, they rely on the in-built explosive charge and the design of the warhead to propel a

KONTAKT-5 ERA

The best example of ERA in the modern age is the Kontakt-5 modules developed by the Russians. This ERA is so good that most NATO forces deem it to be impenetrable to their current AT ammo, including their Uranium-based APFSDS.

needle of super-heated copper through the armour. This means that the effective range of HEAT rounds is from zero to infinite. It is however, only effective against plate armour and low-grade composite armour.

- **HESH:** High Explosive Squash Head ammo hosts a head that's loaded with plastic explosive in a relatively soft casing. On impact, the head flattens against the armoured plate and detonates, causing the inner side of the armour to spall, i.e., unleash a cloud of high-velocity metal shards, flakes from the armour itself, that can prove lethal to crew and equipment. This ammo is useless against spaced armour. 🇷🇺

APPROVED 2013 HARDWARE

BEST OF COMPUTEX 2013 HARDWARE

AWARD WINNER EDITORS' CHOICE SILVER

MUST HAVE

EDITOR'S PICK

ASRock

Get One, The Best One

Z87 Extreme6

M8

Z87 Extreme6/ac

Z87 OC Formula

Z87E-ITX